Fabricating the future

Forget smart watches and augmented reality glasses, the clothes that you wear are the wearables of the future

Arnab Mukherjee
arnab@digit.in

A close up of a moisture wicking fabric showing the pores for capillary action.

The world of fashion knows what we want to wear, often even before we do. Be it the trending hemlines or the latest accessories, designers around the world predict and often create the future when it comes to clothing preferences. And of late, one thing that has been picked up by a lot of such visionary designers is smart fabrics. The connotation 'smart' here isn't just the same as it would be for a smart-watch or a smartTV, but also something more that has been around for a while, especially in the world of sports. But in both cases, they are turning the old adage of 'you are what you wear' on its head.

Performance Fabrics

For most of history, clothing has been a means of preserving one's modesty and hygiene along with projecting one's individuality. But in recent times, that has been changing drastically. Fabric has been developed in the last decade that is specifically engineered to enhance the performance of athletes due to various traits of such fabrics. Some of these traits are achieved quite easily. Take seamless garments as an example. Something that we take for granted, the simple modification of removing seams from sportswear makes them much more adaptable to the physique of athletes.

Taking this a step further, a number of sportswear brands use special blends for their fabric which grants them additional capabilities. Luon from Lululemon is used in their range of trousers and sportswear as the special yarn used in it creates capillary action that draws sweat away from the body. There are multiple other brands that are doing this too. Along with this, compression clothing is also used by a lot of athletes to prevent muscle fatigue and improve blood circulation while reducing muscle recovery time.

The next level in performance enhancement fabrics has been created by BioLogic, a group of researchers from the Tangible Media Group within the MIT media lab by combining bio-materials research with textile design. Instead of using traditional electronics, BioLogic is growing actuators that are alive and responsive bio skin. They do this by harnessing a biological property in a particular species of ancient bacteria, *Bacillus subtilis natto*. The cells of this species expand and contract on the basis of atmospheric moisture, making them a key ingredient in creating a breathable fabric. These cells are harvested in the lab and assembled with a micron resolution bio-printing system to create a second skin-like fabric. This fabric reacts to body temperature to open up flaps around various heat zones allowing sweat to evaporate.

E-textiles

In 2014, Ralph Lauren debuted the tennis collection in the US Open Tennis Championships to be worn by ball boys and girls. The t-shirt could monitor breathing, heart rate and stress. Similar designs and innovations have been seen across the fashion world with one thing in common - the innovation is right down to the yarn level.

Interestingly, conductive yarn has been around for more than a 1000 years. Gold-wrapped threads have been used by artisans for embroidery and fabric construction for a long time. Currently, this skill has been identified as the perfect way to create an electronic canvas in the fabric, removing the need for wires to be embroidered into the fabric. A shining implementation of this is Google ATAP's Project Jacquard which uses conductive yarn.